

EXPERIMENT NO. 40 FLUORESCENCE OF URANYL NITRATE HEXAHYDRATE

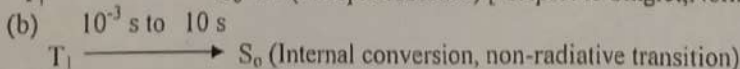
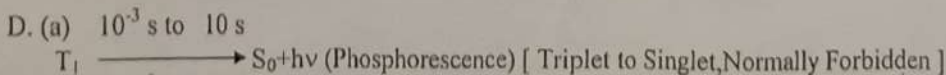
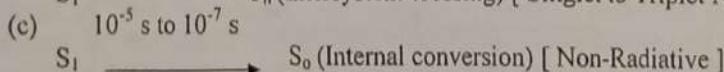
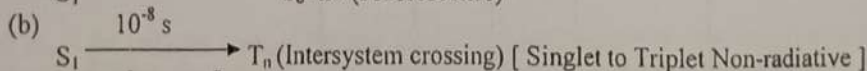
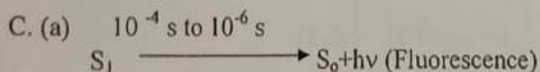
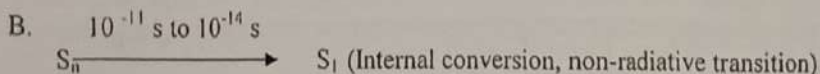
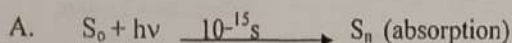
I Objectives

- (i) Calibration of Carl Zeiss monochromator using a mercury source.
- (ii) Recording of the fluorescence spectrum of uranyl nitrate hexahydrate (UNH) excited with an U-V light.
- (iii) Finding the 0-0 electronic transition and associated totally symmetric vibrational frequency in the ground state of the Uranyl ion (UO_2^{++}).
- (iv) To calculate the force constant for the symmetric vibrational frequency and the first electronic excitation energy.

II Definitions

- (a) Luminescence: It is the radiation (photons) emitted by materials by any process except by black body emission. Electroluminescence, chemoluminescence, thermoluminescence, photo-luminescence, etc. are various examples of luminescence.
- (b) Photo-luminescence: In this process radiation is emitted by atoms or molecules excited by photons.
- (c) Fluorescence: It is photo-luminescence with decay time of the order of 10^{-4} sec or less.
- (d) Phosphorescence: It is photoluminescence with decay time of 10^{-3} sec or higher.
- (e) Radiative decay: De-excitation of a system accompanied by emission of photons.
- (f) Non-radiative decay: De-excitation of a system without emission of photons.

Few of the radiative and non-radiative processes mentioned below are shown in Fig.1 (not to scale).



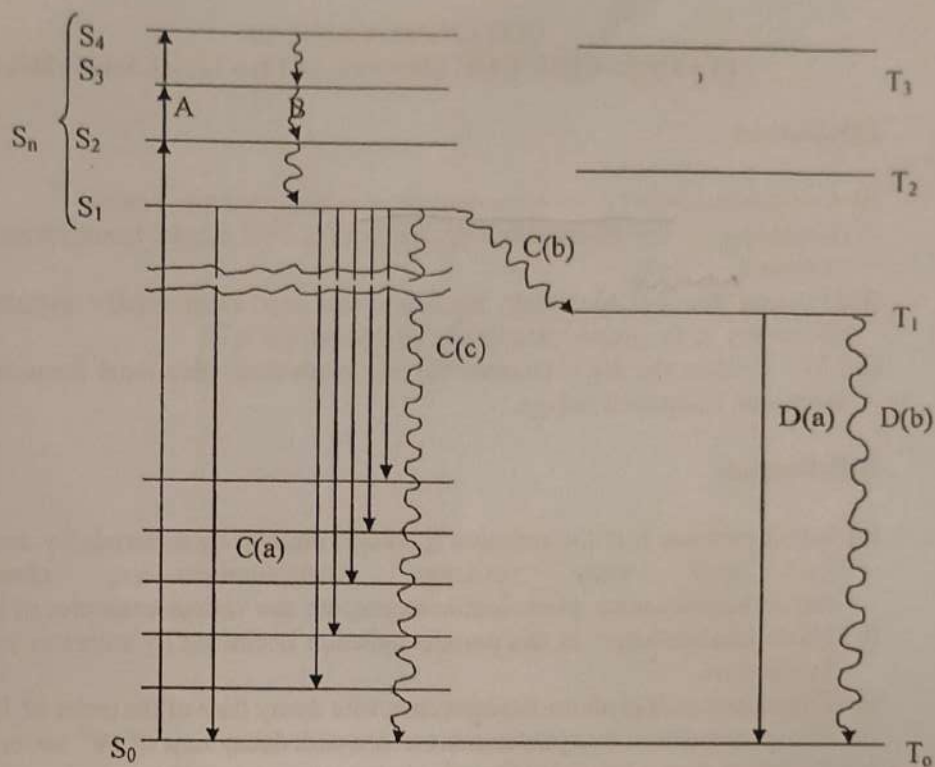


Fig. 1: Some radiative and non-radiative processes.

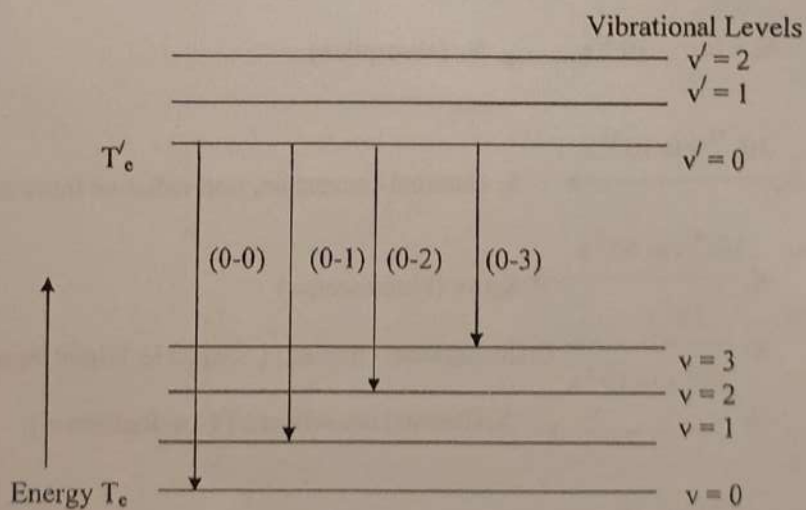


Fig. 2: Fluorescence transitions of UNH.

III Theory

The total energy E_T of the molecules, is the sum of electronic, vibrational and rotational energies.

$$E_T = E_e + E_v + E_r. \quad (1)$$

In the notation of term values

$$T = T_e + G(v) + F(J). \quad (2)$$

Neglecting rotational energy for solid state spectra

$$T = T_e + G(v). \quad (3)$$

Denoting the terms related to the excited state with a prime (/), the transition between excited and ground states are shown in Fig. 2.

The transition frequency ω (cm^{-1}) is given by

$$\begin{aligned} \omega &= T' - T = (T_e' - T_e) + [G'(v') - G(v)] \\ &= \omega_0 + \{ \omega_e'(v' + 1/2) - \omega_e'\chi_e'(v' + 1/2)^2 + \dots \} \\ &\quad - \{ \omega_e(v + 1/2) - \omega_e\chi_e(v + 1/2)^2 + \dots \}. \end{aligned} \quad (4)$$

Neglecting the terms containing anharmonicities χ_e , χ_e' and their higher powers

$$\omega = \omega_0 + \omega_e'v' - \omega_e v. \quad (5)$$

Assuming that transition takes place from $v' = 0$ in solid state

$$\omega = \omega_0 - \omega_e v. \quad (6)$$

It is obvious from Eq. (6) that ω_e is the difference (in cm^{-1}) between successive bands. For calculating ω_e , take the difference between successive bands and average it out. The force constant for the symmetric vibrational frequency can be calculated from the equation

$$\omega_e = \frac{1}{2\pi c} \left(\frac{f}{m_0} \right)^{1/2}, \quad (7)$$

where m_0 = the reduced mass of oxygen atom, f = force constant for the vibration and c is the speed of light in vacuum. The relation is derived under the assumptions of the collinearity of structure of Uranyl ion, harmonicity of vibrations and the applicability of valence force field. If the force constant f is in dyne per cm and m_0 is the reduced mass in the atomic weight units (Aston scale), the simplified form will be

$$f = 5.8883 \times 10^{-2} m_0 \omega_e^{1/2} \text{ dyne/cm.}$$

You will observe at least four fluorescence bands. Make use of Eq. (6) to calculate ω_0 . This is the difference between the electronic energies of the excited and the ground state and is referred as the excitation energy.

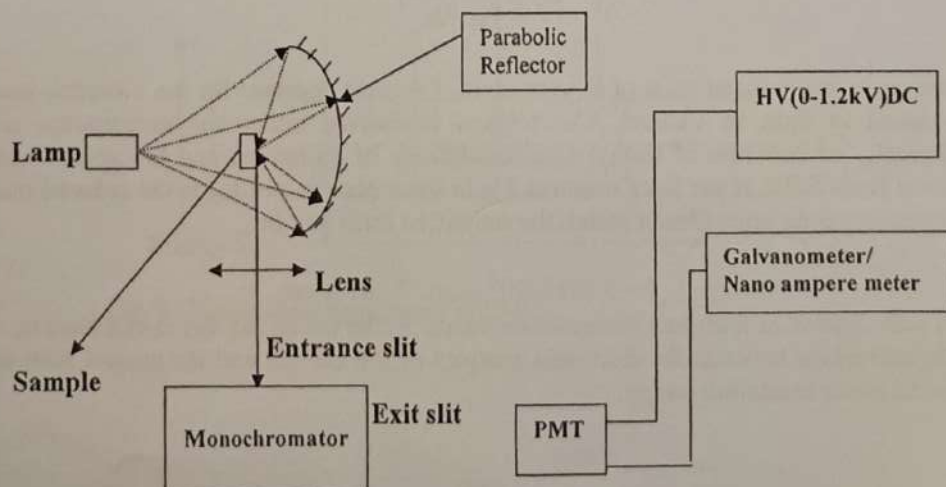
IV Experimental Method

1. Recording of the fluorescence spectrum:

The schematic arrangement for fluorescence set-up is given in Fig 3. The U-V lamp emitting at 365.0 nm is shown in front of the parabolic mirror which focuses the U-V radiation onto the sample. The fluorescence radiation from the sample is focussed by a lens assembly at 90° onto the entrance slit of the monochromator. The light coming out of the monochromator exit slit after dispersion falls on the photo-multiplier tube (PMT) which amplifies the resulting photoelectric current by several orders of magnitude. The output of the PMT is fed to a nano-ampere meter (or galvanometer) for the purpose of recording the signal. With the help of a rotating drum on the monochromator one can explore the presence of the fluorescence bands. Maximize the current in nano-ampere meter (or galvanometer deflection) for the most intense fluorescence band. Both the entrance and the exit slits of the monochromator should be kept less than 2 mm wide for coarse alignment and reduced to 0.5 mm for better resolution. Repeat the experiment several times. Plot a curve for intensity (current in nano-ampere meter /galvanometer deflection) versus monochromator drum reading and locate the fluorescence peaks. (Maximum applied voltage across PMT is 650Volts)

2. Calibration of the monochromator

Irradiate the entrance slit of the monochromator with the mercury vapor lamp instead of the fluorescence light. The ray diagram for the monochromator is shown in Fig 4. Record the current in nano-ampere meter (galvanometer deflection) against the monochromator drum reading similar to part 1 above. Since the intensity of the mercury lamp radiation is much more intense as compared to the fluorescence light the current in nano-ampere meter (galvanometer deflection) may overshoot. Do not reduce the slit width to control the current/ deflection, instead reduce the PMT voltage to bring the current in nano-ampere meter (or maximum galvanometer deflection about 100 div) within the range. Take readings within every measurable fraction of drum reading, especially near the peak of a band. To avoid backlash error, take readings for a unidirectional rotation of the drum head. (Maximum applied voltage across PMT is 300Volts)



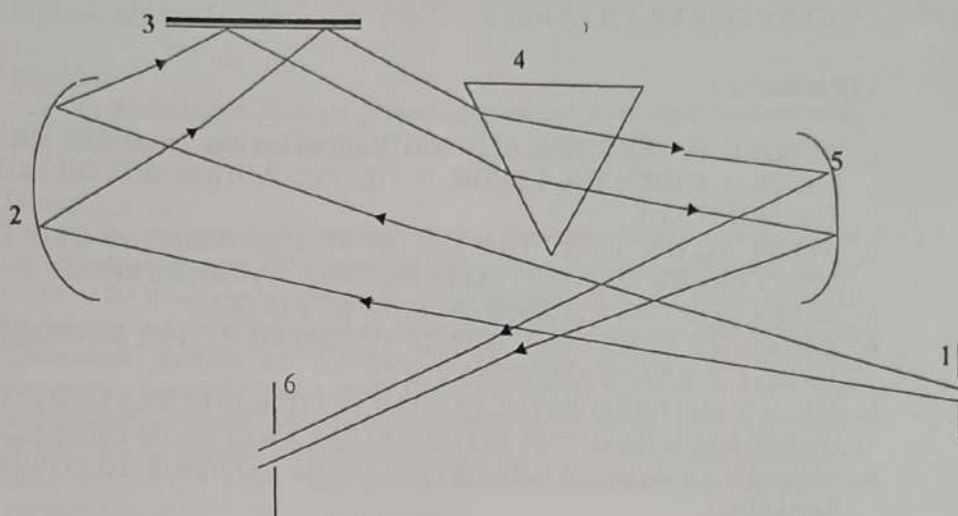


Fig. 4 Ray-diagram for the monochromator.

1. Entrance slit, 2. Reflecting concave mirror, 3. Reflecting plane mirror, 4. Prism, 5. Reflecting concave mirror, and 6. Exit slit.

V Analysis

- Plot the intensity (galvanometer deflection) versus the drum reading for the mercury lamp.
- Identify the wavelength of the intense peaks appearing in the above plot from Table I and then plot a graph between wavelength and drum reading for calibration.
- Convert the intensity (galvanometer deflection) vs. drum reading plot for fluorescence spectrum into an intensity vs. wavelength plot using the calibration curve.
- Analyze the spectrum by identifying the fluorescence band positions (in nm) and obtain the information required in objectives (iii) and (iv) above.

Table I: Emission spectrum of mercury vapour lamp

Wavelength (nm)	Colour	Relative Intensity
404.7	Violet (Deep)	1800
407.8	Violet	150
435.8	Indigo - Violet	4000
491.6	Blue-green (Deep- Green)	80
546.1	Green (yellowish)	1100
577.0	Yellow-orange	240
579.1	Yellow-orange	280
623.4	Red	
690.8		

VI Precautions

1. NEVER KEEP VERY WIDE SLIT WIDTH (Keep less than 2 mm). THE SMALLER IS THE SLIT WIDTH THE BETTER IS THE RESOLUTION BUT DO NOT CLOSE THE SLIT FULLY.
2. NEVER ALLOW LIGHT TO FALL ON TO PMT DIRECTLY. PREVENT ALL STRAY LIGHTS BECAUSE IT MAY PRODUCE SPURIOUS PEAKS.
3. NEVER KEEP PMT VOLTAGE ON WHEN NOT IN USE.
4. THE POWER SUPPLY CAN GIVE UP TO 1200 VOLTS. BUT BE CAREFUL NOT TO GIVE VOLTAGE ABOVE 700 VOLTS.
5. THE VISIBLE LIGHT IN THE PRESENT SETTING IS IN THE RANGE 2400-2850 OF THE DRUM READINGS. DO NOT GO FAR AWAY FROM THESE READINGS.
6. ALWAYS MEASURE FROM HIGHER DRUM READINGS TO LOWER DRUM READINGS.
7. U-V RADIATIONS FROM U-V LAMP ARE HARMFUL TO YOUR EYES. DO NOT LOOK INTO THE LAMP DIRECTLY.

VII Questions

1. Why do we use a metallic parabolic reflector in this experiment?
2. What is the source of generation of U-V light in the U-V lamp in this experiment?
3. Comment on the functional dependence of drum reading vs. wavelength? Would it be the same or different if we were to use a grating for dispersion?
4. What material is the optics made of? Why do we gather fluorescence radiation at 90° geometry?
5. How would the intensity of the fluorescence changes if the temperature of the salt is increased or decreased?
6. If the sample UNH is dissolved in water, will it still produce the same fluorescence spectrum?

VIII References

1. R. S. Becker, "Theory and Interpretation of Fluorescence and Phosphorescence."
2. G. M. Barrow, "The Structure of Molecules."
3. G. Herzberg, "Spectra of Diatomic Molecules," Ch. 4.
4. C.N. Banwell, "Fundamentals of Molecular Spectroscopy."